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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

II Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2023

MA1003-1 - DIFFERENTIAL EQUATIONS & LAPLACE TRANSFORMS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Solve the differential equation $\frac{d^2y}{dx^2} + 4y = 2e^x \cos 2x$	5	L*1	2	1
b)	Using the method of Variation of parameters solve the differential equation $(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$	5	L2	2	1
2. a)	Solve the differential equation $x^2y'' + xy' + y = 2 \cos^2(\log x)$	5	L2	2	1
b)	Solve $(D^2 - 1)y = (1 + x^2)e^x$	5	L1	2	1
Unit – II					
3. a)	If $L\{f(t)\} = F(s)$ then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$.	5	L3	3	1
b)	Rewrite the following function using unit step function and find its Laplace transform: $f(t) = \begin{cases} \sin t & \text{if } 0 \leq t < \pi/2 \\ \cos t & \text{if } t \geq \pi/2 \end{cases}$	5	L2	3	2
4. a)	Find the Laplace transform of $e^{-3t} \sin 5t \sin 3t$.	5	L2	3	2
b)	Find the Laplace transform of $\int_0^t \frac{\cos 3t - \cos 5t}{t} dt$.	5	L2	3	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
